RIS-Aided Integrated Sensing and Computing Communication: Frontier Advances and Prospects

Xuejie Hu, Yue Tian, Jin Huang, Xianling Wang, and Haijun Zhang

Abstract

With the rise of a new wave of technological and industrial revolutions, integration scenarios based on the convergence of communication, sensing, and computing are driving the innovative applications and rapid expansion of wireless communication networks and their associated vertical industries. As a means of lossless signal reflection, reconfigurable intelligent surfaces (RISs) define a new wireless transmission paradigm with its exceptional channel control capabilities. In particular, RIS-assisted semantic communication and integrated communication-sensing systems enable multi-dimensional, in-depth applications, gradually becoming a dominant trend in sixth-generation (6G) intelligent communication and sensing technologies. This article focuses on the technological advantages of RIS in current integrated scenarios and its application prospects in future communication networks. It introduces cutting-edge research advancements and forecasts in RIS-assisted communication, sensing, and computing integration, highlighting security conflict avoidance methods for sensitive information scenarios, joint optimization methods for RIS-based multimodal information fusion, and intelligent algorithm adaptation mechanisms for RIS-assisted communication-sensing systems. Furthermore, the article provides the latest research support and innovative insights into the cross-dimensional information interaction mechanisms within RIS-assisted integrated information networks and scenarios, addressing current challenges and offering future outlooks.

Introduction

In fifth-generation (5G) networks, large signal bandwidth, massive connections, machine learning capabilities, and high-directionality signal transmission have further enhanced the performance of cellular networks. Currently, both academia and industry are actively exploring more advanced mobile cellular network positioning technologies and intelligent algorithms, aiming to equip sixth-generation (6G) with native support for communication and sensing applications [1], while achieving deep integration between the digital world and communication physical layer networks. Reconfigurable intelligent surface (RIS), as an innovative technology with intelligent reflection and tuning capabilities, has potential applications in overcoming coverage holes, enabling high-precision positioning, ensuring secure communication, and reducing latency in mobile edge networks [2]. Compared to traditional communication systems, RIS not only improves signal coverage and transmission efficiency but also enables precise sensing and data collection in target areas through flexible beamforming techniques. This makes RIS a key driving force behind the integration of communication, sensing, and computing technologies, represented by 6G intelligent communication-sensing integration technology and semantic communication (SemCom) systems [3].

Currently, the main challenge in the information fusion scenarios of wireless communication and SemCom system is achieving the secure, efficient, and robust transmission of semantic information under the constraints of semantic distortion. The semantic channel model, joint semantic-channel coding (JSCC), and encryption coding schemes in RIS-assisted SemCom systems, based on information path optimization, are essential for addressing demand-oriented semantic tasks [4, 5]. Additionally, RIS can integrate multimodal information content to form deep semantic perception, providing reliable data support for environmental reconstruction in communication-sensing-computing fusion scenarios.

RIS, with its exceptional channel control capabilities, provides new possibilities for the integration of communication, sensing, and computing in future 6G and beyond wireless networks, promoting the development of intelligent communication systems toward greater efficiency and integration. In applications where sensing and communication are deeply fused, RIS can support multidimensional information transmission and processing, covering various scenarios such as intelligent transportation, industrial internet of things (IIoT), and smart homes. Furthermore, the introduction of RIS technology offers new methods for addressing privacy and security issues in sensitive perception information scenarios, providing fresh perspectives for cross-domain resource collaboration optimization in multimodal information fusion settings. With the aid of intelligent decision-making algorithms, RIS-assisted com-

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Digital Object Identifier: 10.1109/IOTM.001.2400168

2576-3180/25/$25.00 © 2025 IEEE

munication and sensing positioning technologies are now expected to meet lower latency and higher centimeter-level positioning accuracy requirements. As the demand for efficient data transmission, precise acquisition of multidimensional information, and intelligent decision-making continues to grow, more innovative research methods are needed to better drive the evolution from the internet of things (IoT) to an intelligent internet of everything [6].

Based on the latest research advancements in RIS-assisted information communication-sensing-computing fusion scenarios, this magazine article presents new insights into information security protection strategies, multimodal information joint optimization methods, and intelligent algorithm-assisted information interaction mechanisms. The specific contributions are as follows:

- From the perspective of privacy information transmission in SemCom systems and security path optimization based on perception coverage vulnerabilities, this article discusses methods for avoiding security conflicts and information transmission mechanisms in sensitive information scenarios assisted by RIS.
- Proposes an advanced network architecture design for RIS-assisted multimodal information fusion scenarios, including SemCom path optimization and a cross-dimensional collaborative optimization framework for integrated communication and sensing. Additionally, it offers targeted insights into efficient resource allocation methods in the context of quantum computing.
- Highlights innovative algorithm designs for RIS-assisted communication and localization integration tailored for smart service applications, along with low-latency localization strategies for RIS-assisted communication-sensing integration. Additionally, it proposes a RIS-assisted semantic channel selection method based on user quality of experience (QoE) and quality of service (QoS).

Privacy Security Protection Method of RIS Auxiliary Sensitive Information Scenarios

This section primarily investigates how to leverage RIS-assisted SemCom systems to enhance information security in semantic communication-sensing scenarios that prevent privacy information leakage as shown in Fig. 1. It explores the identification of perception coverage vulnerabilities and the development of security path optimization algorithms to establish a sensitive information transmission mechanism for integrated communication-sensing-computing scenarios, thereby achieving eavesdropping avoidance and information security protection for the system.

Privacy Protection of SemCom Systems

Since SemCom system transmits a compressed amount of compressed information, the security of semantic information transmission becomes critically important. Enhancing information security strategies in SemCom systems with simultaneously transmitting and reflecting (STAR)-RIS features [7], developing RIS-based methods for secure channel selection of user-sensitive semantic information, and establishing privacy protection frameworks have become hot research topics.

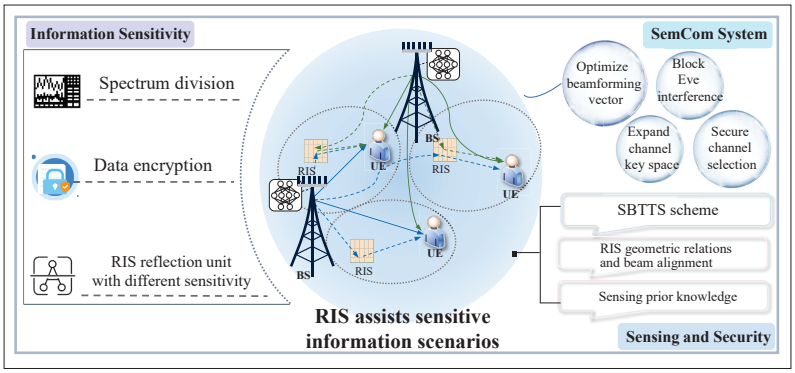

FIGURE 1. RIS auxiliary sensitive information scene structure diagram.

In contrast to traditional RIS, STAR-RIS supports channel partitioning by dividing its surface into distinct regions for simultaneous transmission and reflection, allowing different users or devices to access unique transmission paths. STAR-RIS-assisted SemCom systems enable channel isolation between legitimate users and potential eavesdroppers through dynamic beamforming and channel partitioning, which blocks the transmission of semantic signals between legitimate users and the BS, thus enhancing communication security. Additionally, the system utilizes a random switching sequence of RIS-assisted channels and introduces a correlation-based semantic channel estimation scheme tailored for legitimate users. For channel prediction, an eavesdropper channel simulation-based generative adversarial network (ECS-GAN) is employed, accounting for both the sensitivity of semantic transmission symbols and the channel's overall secure capacity. A secure membership function generated through fuzzy logic decision-making guides secure channel selection in the RIS-assisted SemCom system, achieving effective privacy protection in semantic communication systems.

For future wireless network scenarios that demand high throughput, such as broadband communication, the current limitation on key generation rates presents a challenge. RIS can maximize channel cognition and further adapt channels as needed, expanding the key space for physical-layer key generation. As a result, it can serve as one of the methods to enhance the key generation rate and protect the privacy and security of SemCom systems. Meanwhile, by leveraging semantic encoding and semantic representation in the SemCom system, combined with RIS-based encryption and authentication, and integrating with traditional cryptographic authentication protocols, a lightweight security enhancement mechanism can be designed to safeguard the privacy of SemCom systems.

Perceptual Coverage and Secure Path Optimization

In integrated communication-sensing systems, the competition for resources between communication channels and sensing channels can lead to perception coverage vulnerabilities, affecting the system's ability to accurately detect and respond to the surrounding environment [8–10]. For instance, in autonomous driving scenarios, resource contention may prevent radar or sensing devices from timely detecting obstacles, potentially resulting in safety incidents.

Numerous current studies are focused on developing intelligent and secure path planning systems that can automatically adjust using RIS in dynamic environments.

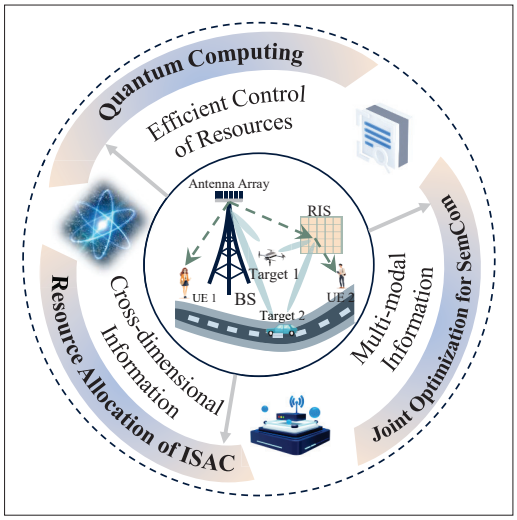

FIGURE 2. RIS-assisted multi-modal information fusion scene structure diagram.

The simultaneous beam training and target sensing (SBTTS) scheme based on RIS-assisted systems can utilize the mixed echoes from RIS and targets to differentiate between sensing targets. Additionally, the geometric relationships facilitated by RIS enable beam alignment in integrated sensing and communication (ISAC) systems, effectively mitigating the accuracy and timeliness issues in target detection caused by perception coverage vulnerabilities.

STAR-RIS and intelligent omni-surfaces (IOS), with their capability for 360-degree omnidirectional coverage, can significantly enhance multi-node wireless sensing data collection. Their multi-path diversity and channel isolation capabilities allow for more accurate differentiation of data transmission paths for different users or sensing devices. Additionally, the phase data of transmitted signals can be used as prior knowledge to adaptively adjust the coverage of the Fresnel sensing zone, thereby improving the system's secure capacity.

Numerous current studies are focused on developing intelligent and secure path planning systems that can automatically adjust using RIS in dynamic environments. These approaches construct a flexible signal reflection phase adjustment model for RIS based on prior data within complex communication-sensing environments. It assesses the degree of interference affecting the physical layer channel between users and the BS based on the overall security capacity of the communication-sensing network, allowing for real-time adaptation to changing network and physical environments, thereby ensuring the coverage breadth and security of communication paths within the ISAC system.

Sensitive Information Reconstruction and Transfer Mechanism

The research focused on preventing eavesdropping and enhancing user privacy protection in communication-sensing fusion scenarios, the precise categorization of sensitive information and the selective transmission of secure channels are crucial for preventing potential eavesdroppers from stealing sensitive information and ensuring user data security. Utilizing different hardware design methods for RIS, such as placing *L*-shaped active sensors behind the transmission elements of RIS to achieve reverse semantic perception in wireless sensing, allows multiple signal spectra to be encoded into a single metaspectrum. Sampling methods can then be employed to select the signal spectra related to sensitive information categorization tasks, enabling the reconstruction of sensitive information [11]. In addition, utilizing the dynamic amplitude response matrix sequence of the RIS as a type of key, and by randomizing amplitude and phase responses on the RIS, enables random control of the signal paths, facilitating the covert transmission of sensitive information.

In multi-user and multi-task scenarios, it is necessary to balance channel resource allocation between different users and tasks to meet the transmission requirements of communication information and sensitive sensing information. Each reflective element of the RIS can independently adjust its phase, exhibiting different sensitivities to the incoming signals from different tasks and users from various directions. By analyzing the signal characteristics received by each reflecting element of the RIS, a channel knowledge map with measured radio parameters and channel resource allocation information can be obtained. This map can then be used to derive a channel matching scheme tailored to the requirements of sensitive information transmission.

Network Architecture Design of Multi-Modal Information Fusion Scenario Based on RIS

This section primarily introduces the joint optimization methods for RIS-assisted SemCom systems in multimodal semantic information fusion scenarios. It covers the multidimensional collaborative optimization methods for communication information and sensing information within integrated communication-sensing systems, as well as the optimization methods for efficient communication resource allocation using shared RIS-assisted communication resources in the context of quantum computing as shown in Fig. 2.

Joint Optimization Strategy of Multimodal Information in Communication Semantic Fusion Systems

Research on RIS in SemCom primarily focuses on enhancing the adaptability of communication systems. Compared to traditional point-to-point SemCom systems, the resource-efficient management strategy of RIS-assisted SemCom systems highlights the potential of RIS to improve semantic transmission efficiency and system robustness in practical applications. SemCom emphasizes the essence or meaning of information transfer, and RIS can dynamically control the selection of channels that are most conducive to conveying semantic information, optimizing the information entropy in SemCom. This enables more effective transmission of meaningful data while preserving the integrity and contextual continuity of semantic information. The use of RIS-assisted channels in SemCom systems enhances channel selectivity, allowing both semantic and non-semantic users within the entire communication system to dynamically allocate resources, maximizing the overall user experience for all users. In addition to enhancing signal quality in specific directions to reduce semantic errors, RIS can support higher

user density and enable large-scale semantic transmission systems through dynamic resource allocation across different reflecting elements.

In response to the challenges faced by SemCom systems and semantic channel coding technologies in the context of communication intelligence and information fusion scenarios, [3] proposes a mutual information (MI)-based joint optimization strategy for RIS-assisted SemCom, as shown in Fig. 3. As shown in Fig. 3, the RIS-assisted SemCom system provides a substantial gain in MI, enhancing the transmission of compressed information and improving decoding and recovery performance. By calculating the MI for both RIS-assisted communication transmission scenarios and RIS-assisted SemCom transmission scenarios, the strategy allocates weighting coefficients to these two scenarios. Under the constraints of minimum signal-to-noise ratio and minimum transmission rate, it derives the optimal solution for MI. This optimal solution is then utilized to develop multi-user channel selection and allocation schemes for RIS-assisted SemCom systems.

Interdimensional Collaborative Resource Allocation Framework of Synaesthesia Integration

The cross-dimensional collaborative transmission of communication and sensing information can not only effectively improve resource utilization but also ensure the independent transmission of sensitive sensing and communication data, thus preventing third-party eavesdropping and enhancing network security. To strengthen network resilience and prevent the information coupling risks arising from the cross-transmission of integrated sensing and communication data, a coding scheme is proposed for filtering and reconstructing sensitive sensing and communication information. This approach enhances overall network security and safeguards user data privacy, effectively increasing privacy protection during the collaborative transmission of sensing and communication information.

Recently, much research [12, 13] has focused on joint optimization methods for enhancing target detection probability and communication coverage in RIS-assisted ISAC systems. The goal is to maximize the signal-to-noise ratio (SNR) for communication users while ensuring that the threshold detection probability for sensing is met, enabling cross-dimensional resource coordination between communication and sensing. Given that optimizing the performance of sensing targets and communication users often presents conflicting requirements, the approach involves jointly optimizing and designing the transmit precoding matrix and the RIS phase shift matrix. This includes considerations for the design of signal receiving filters for target detection and the reflection coefficients of the RIS, ensuring that the overall rate for communication users is maximized while satisfying the success rate requirements for target detection.

Efficient Control of Resources in Quantum Computing

Current quantum networks have unlocked new applications in areas such as quantum sensing, distributed quantum computing, and quantum communication protocols. However, compared to classical networks, quantum architectures face

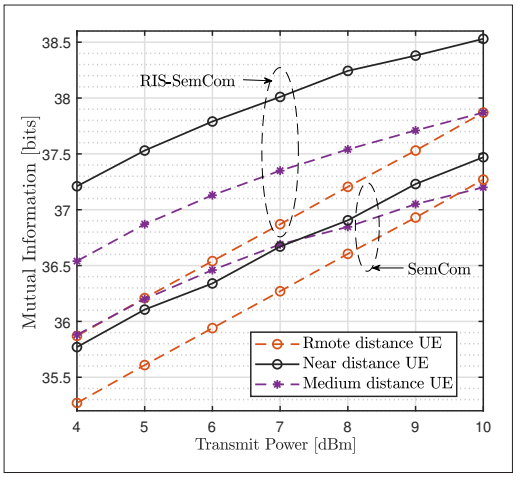

FIGURE 3. Transmit power versus MI in RIS-SemCom system and SemCom system.

deployment challenges due to their reliance on fiber channels to share quantum information, particularly in regions with obstacles and limited infrastructure. Recent research on free-space optical (FSO) quantum networks (QN) [12] highlights challenges such as the need for clear line-of-sight (LoS) between communication nodes and fair resource allocation among users with heterogeneous quality of service (QoS) requirements. Since RIS has the potential to establish virtual LoS links between communication nodes without requiring quantum storage and offers significant advantages in energy efficiency, the integration of RIS with quantum links has been shown in previous studies to alleviate the LoS requirements of FSO systems. In addition, the combination of the quantum approximate optimization algorithm (QAOA) and quantum computing enables fast and efficient solutions to complex combinatorial optimization problems in resource allocation for RIS-assisted wireless communication systems.

The high channel correlation of signals transmitted by multi-antenna arrays on RIS in low-frequency communication systems weakens channel independence, causing inter-channel interference that reduces overall system gain. A significant challenge currently faced, and a focal point for future solutions, lies in effectively coordinating resource allocation and enhancing overall system performance for RIS-assisted multi-user, multi-cell communication within low-frequency networks. To address this, QAOA is proposed to partition received signals on the RIS side and assign them to different RIS reflection units. By selecting more distant reflection units, this approach aims to increase channel independence, reduce inter-channel correlation, and enhance system gain.

RIS Auxiliary Synaesthesia System Intelligent Services Application System

The RIS-assisted ISAC system has achieved seamless integration of communication and sensing functions in intelligent services applications, ensuring efficient cross-layer resource management. Applications in scenarios such as smart cities, autonomous driving, and next-generation wireless networks have positioned RIS as a key driving force for future intelligent systems, fos-

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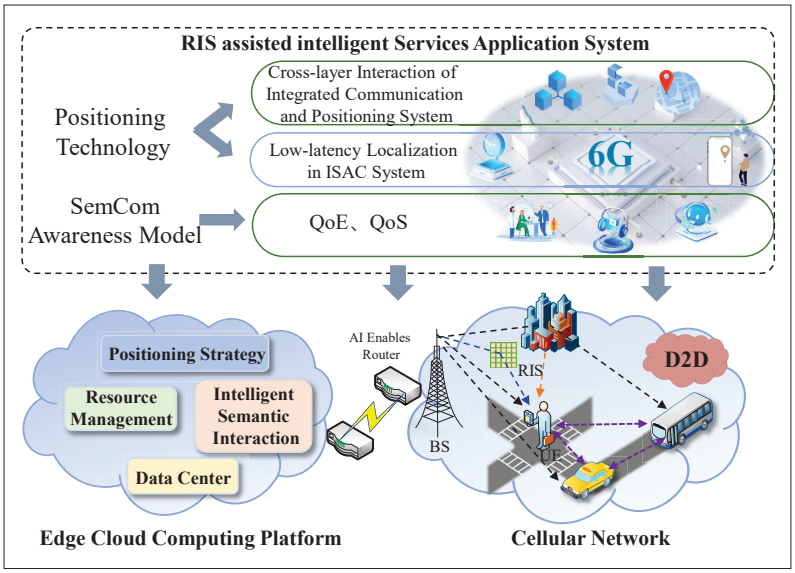

FIGURE 4. RIS assisted intelligent service application systems structure diagram.

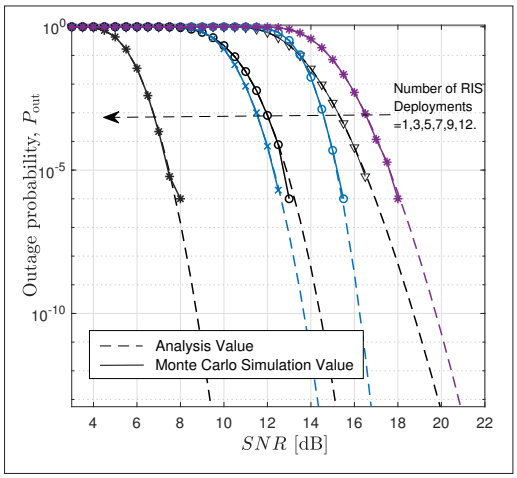

FIGURE 5. SNR versus outage probability for deploying different numbers of RIS.

tering innovation in areas like human-machine interaction, smart homes, and augmented reality. This chapter discusses the application directions of RIS-assisted intelligent business in future wireless communication networks, focusing on the design of cross-domain interactions in communication and positioning systems as shown in Fig. 4, the analysis of low-latency positioning strategies in ISAC systems, and the establishment of user semantic perception models.

Cross-Layer Interaction Design of Integrated Communication and Positioning Systems

In an integrated communication and positioning system, cross-layer interaction refers to the coordination and information sharing between different layers of the network protocol stack, such as the physical layer, medium access control (MAC) layer, network layer, and application layer. Typically, these layers operate independently, but in a cross-layer design, they share resource management information and position estimation data, making decisions based on the overall optimization needs of the system. With the evolution of 6G wireless communication systems, traditional positioning technologies are constrained by the spatial and angular resolution capabilities of wireless transceivers. This limitation prevents conventional devices from accurately distinguishing electromagnetic signals and extracting wireless signal features, making precise and robust sensing and positioning unachievable. The characteristics of high-frequency communication, such as increased signal attenuation and more pronounced multipath effects, place higher demands on line-of-sight (LoS) paths and beam alignment. Additionally, cross-layer interactive positioning methods require feedback of physical-layer channel characteristics to the network layer, which imposes higher requirements on system processing speed and feedback mechanisms. In RIS-assisted integrated communication and positioning systems, statistical analysis and machine learning techniques enable RIS to understand human behavior or intent within wireless networks. This allows the wireless communication system to optimize resource scheduling, making it more intelligent while also enabling RIS to better perceive and recognize human movements. As a result, it can enhance human-computer interaction, virtual tactile reality, and services like smart homes and intelligent healthcare.

Currently, RIS-assisted wireless communication positioning systems employ intelligent edge computing methods, integrating cross-layer resource allocation and optimization of communication and positioning information transmission. This approach effectively reduces the computational burden on central servers while enhancing the overall response speed and reliability of the system. However, for real-time positioning systems in large-scale user scenarios and complex environments, further improving the computational speed and efficiency of algorithms to meet the demands of various real-time communication applications remains one of the challenges for RIS-assisted communication positioning systems [13]. Adaptive RIS control and positioning strategies based on CSI and user behavior data, with autonomous learning and adjustment capabilities, can integrate joint optimization across the physical layer, MAC layer, and network layer. This enables the simultaneous optimization of positioning accuracy and the realization of a more flexible and real-time communication and positioning system.

Synaesthesia Integrated Low Latency Positioning Strategies

As the positioning accuracy and coverage of integrated communication-sensing systems continue to improve, the latency of positioning also imposes higher demands on algorithm processing times. Different positioning scenarios have varying latency requirements, e.i., the positioning latency in vehicular networks needs to be within 1 ms. Currently, RIS-assisted multi-agent collaborative networks have achieved significant performance improvements, as shown in Fig. 5, which shows the performance comparison of a multi-point cooperative system based on different numbers of RIS. Compared to a single fixed RIS node, multi-point assistance significantly reduces the outage probability (OP) of system. Applied to multi-node cooperative positioning systems, node activation strategies and multimodal information fusion

methods effectively reduce positioning delays. This enables agents to obtain relative positioning information, allowing the positioning system to not only achieve centimeter-level accuracy but also enhance positioning efficiency and reduce latency.

In addition to the RIS-based collaborative positioning strategy with multiple intelligent nodes, improving the computational efficiency and complexity of algorithms is also one of the methods for achieving low-latency positioning. By deploying RIS at different network nodes and utilizing it for local decision-making and data processing, reliance on global information and the delays caused by parameter estimation in complex multipath environments can be reduced, thereby enhancing real-time performance. By utilizing virtual line-of-sight (VLoS) delays and direct path delays, combined with the angle of signal transmission, channel parameters can be roughly estimated using inverse fast fourier transform (IFFT). Subsequently, the parameters can be refined using the quasi-Newton method, which significantly reduces the required computational load, achieving the goal of fast signal processing in the frequency domain and reducing computation time [14].

A Model of User Semantic Perception Based on Information Quantification

The method based on information quantification of semantic communication system aims to combine the MI, semantic transmission rate and semantic understanding rate in semantic communication system, and use specific multivariate estimation and quantification strategy to reduce the transmission of redundant information, while ensuring the accuracy and relevance of perceptual information. Based on the quantized MI parameters and Logit regression model, the user's QoE model is established as the user feedback index in the semantic communication system. This index also takes into account the growth rate, scores for semantic rate and semantic accuracy to evaluate the reliability of system transmission.

For the future intelligent business applications of RIS-assisted SemCom systems [15], leveraging machine learning and big data analytics, along with RIS-based edge computing and distributed processing, is essential for designing intelligent QoE and QoS parameters as user feedback metrics. This approach can analyze user behavior and network data to enable real-time adjustments in resource allocation strategies, content caching models, and network traffic management, helping to prevent congestion and data loss and increase user satisfaction. Given that SemCom transmits only a limited amount of compressed information, enhancing the continuity and integrity of information in SemCom systems through RIS is crucial. Innovative compression and encryption techniques focused on user experience perception of semantic information will also be important future research directions.

To further validate the multidimensional perceptual metrics of RIS-assisted SemCom, [3] employed deep learning based SemCom system (DeepSC) to establish a QoE perception model for RIS-based SemCom system. This model balances the semantic understanding rate and semantic transmission rate for text and image transmission within the SemCom system. By utilizing RIS configurations and selecting QoE parameters, the model identifies the optimal channel paths for compressed semantic information transmission, confirming the significant role of RIS in enhancing semantic performance.

Challenges and Prospects

In RIS-assisted synaesthesia fusion scenarios, although RIS technology provides significant advantages for integrated communication and sensing systems, as well as SemCom systems, it also faces several challenges. In multi-user, multi-task environments, channel resource allocation, spectrum sharing, and time-frequency resource scheduling require cross-layer coordination, involving extensive computation and control, which increases the complexity of system design and implementation. In addition, the controllability of RIS enables eavesdropper to use RIS to interfere with communication and transmission of sensing information, which brings challenges to the physical layer security of RIS auxiliary synsensing computing networks. Currently, RIS technology is still in a relatively early stage of development, and manufacturing and design costs remain high due to limited production scale.

RIS is expected to drive many new applications in future communications. In smart cities where wireless networks have blind spots, multi-RIS-assisted SemCom systems and ISAC systems can intelligently adjust signal propagation paths to ensure efficient transmission and understanding of information. In a high-speed mobile environment, optimizing the exchange of information between vehicles and between vehicles and infrastructure can improve traffic safety and efficiency. Currently, privacy and covert communication are rapidly developing in multiple research fields. In RIS-assisted ISAC fusion scenarios, leveraging intentionally induced uncertainties, such as channel condition variations, location uncertainty, and artificial noise. Therefore, leveraging RIS to generate dynamic interference signals to weaken the detection capability of potential adversaries will become a promising approach for protecting sensitive information in the future. With advancements in RIS-assisted wireless technologies, the use of active RIS to distribute computing resources and tasks to improve overall system computational efficiency, while meeting the timeliness requirements of information exchange in communication scenarios, is an issue to be addressed. It should be noted that while RIS applications are vast, the practical deployment of RIS in networks will require careful consideration of system design and optimization, as well as addressing challenges and limitations posed by real-world environments.

RIS is expected to drive many new applications in future communications. In smart cities where wireless networks have blind spots, multi-RIS-assisted SemCom systems and ISAC systems can intelligently adjust signal propagation paths to ensure efficient transmission and understanding of information.

Conclusion

This magazine article presented cutting-edge research in RIS-assisted information communication-sensing fusion scenarios and proposes several innovative methods and strategies aimed at enhancing information security, optimizing multimodal information fusion, and improving the auxiliary effectiveness of intelligent algorithms. First, it addressed the issue of privacy information transmission in SemCom systems, proposing solutions for security conflict avoidance from the perspectives of secure path optimization and perception

coverage vulnerabilities. Next, it designed RIS-assisted multimodal information fusion network architecture, focusing on cross-dimensional collaborative optimization between SemCom and integrated communication-sensing, as well as insights into achieving efficient resource allocation in the context of quantum computing. Finally, based on intelligent algorithm-assisted information interaction mechanisms, this article introduced innovative algorithms for communication and localization integration, low-latency localization strategies, and MI-based RIS semantic channel selection method.

Acknowledgment

This work was supported in part by the Natural Science Foundation of China under Grant 62201482; in part by the Natural Science Foundation of Xiamen under Grant 3502Z20227213; in part by the Natural Science Foundation of Fujian Province, China, under Grant 2023J011455.

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